

```
In [ ]: from ultralytics import YOLO
import cv2
import os
from IPython.display import Video, display
```

```
In [7]: video_path = 'peking_university.mp4' # 输入视频
output_dir = 'frames' # 输出帧目录
os.makedirs(output_dir, exist_ok=True)

cap = cv2.VideoCapture(video_path)
fps = cap.get(cv2.CAP_PROP_FPS)
frame_count = int(cap.get(cv2.CAP_PROP_FRAME_COUNT))
print("视频 FPS:", fps)
print("总帧数:", frame_count)

frame_id = 0
while True:
    ret, frame = cap.read()
    if not ret:
        break
    cv2.imwrite(f"{output_dir}/frame_{frame_id:05d}.jpg", frame)
    frame_id += 1
cap.release()

print(f"已切分 {frame_id} 张图片")
```

视频 FPS: 23.987918449534355

总帧数: 3971

已切分 3971 张图片

切分的图片数量与视频总帧数一致，已成功提取所有视频

```
In [7]: f = open('./datasets/labels/train/frame_00000.txt')
lines = f.readlines()
for line in lines:
    print(line)
```

```
0 0.802083 0.710478 0.137500 0.560662

0 0.347396 0.602941 0.048958 0.161765

1 0.750000 0.555147 0.047917 0.058824

2 0.350521 0.665441 0.073958 0.128676

0 0.407292 0.597426 0.045833 0.117647

0 0.398438 0.595588 0.028125 0.117647

3 0.966667 0.656250 0.064583 0.099265

0 0.418750 0.596507 0.029167 0.104779

0 0.419792 0.586397 0.025000 0.084559

0 0.413542 0.599265 0.037500 0.113971

2 0.342187 0.679228 0.057292 0.104779

0 0.846354 0.683824 0.071875 0.492647
```

图片的目标框坐标分别为：类别标签id、目标框的归一化中心坐标位置
(x,y) , 目标框的归一化宽和高 (w,h)

```
In [9]: import os
print(os.listdir('./datasets/'))
print(os.listdir('./datasets/images'))
print(os.listdir('./datasets/labels'))

['images', 'labels']
['train', 'val']
['train', 'val', 'train.cache', 'val.cache']
```

```
In [12]: import os
import cv2
import glob
import random
```

```

import matplotlib.pyplot as plt

names = {
    0: "person",
    1: "bicycle",
    2: "car",
    3: "motorcycle"
}

image_dir = "frames"
label_dir = "labels"

all_images = glob.glob(os.path.join(image_dir, '*.jpg'))
sample_images = random.sample(all_images, 5)

for img_path in sample_images:
    txt_path = img_path.replace(image_dir, label_dir).replace('.jpg', '.txt')

    image = cv2.imread(img_path)
    h, w, _ = image.shape

    if os.path.exists(txt_path):
        with open(txt_path, 'r') as f:
            for line in f:
                cls, xc, yc, bw, bh = map(float, line.split())

                # YOLO → 像素坐标
                x1 = int((xc - bw/2) * w)
                y1 = int((yc - bh/2) * h)
                x2 = int((xc + bw/2) * w)
                y2 = int((yc + bh/2) * h)

                # 画框
                cv2.rectangle(image, (x1, y1), (x2, y2), (0,0,255), 2)

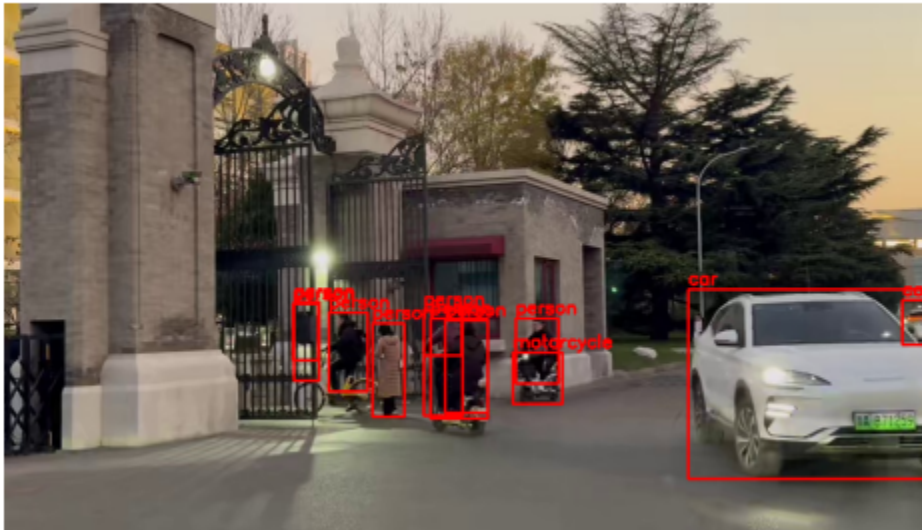
                # 类别名称
                cls_name = names.get(int(cls), str(int(cls)))
                cv2.putText(image, cls_name, (x1, y1 - 5),
                           cv2.FONT_HERSHEY_SIMPLEX, 0.6, (0,0,255), 2)

# 显示可视化结果
plt.figure(figsize=(6,4))

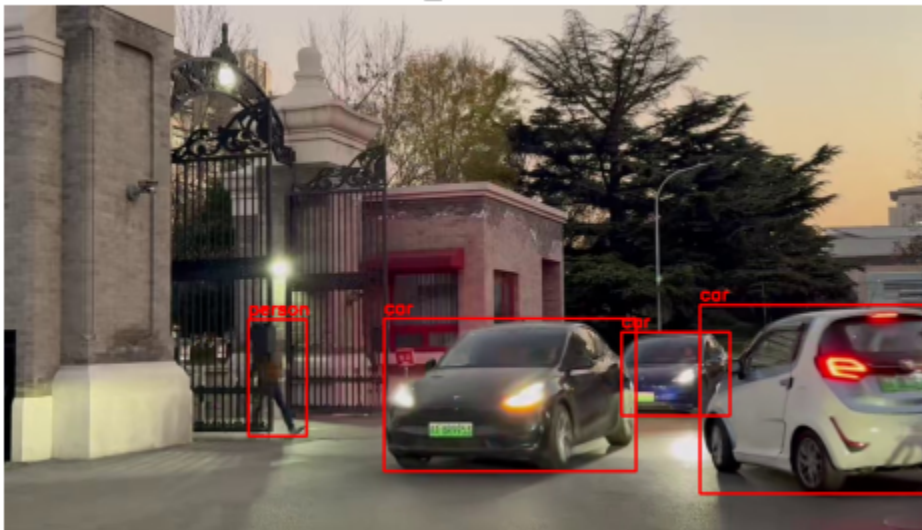
```

```
plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))  
plt.title(os.path.basename(img_path))  
plt.axis('off')  
plt.show()
```

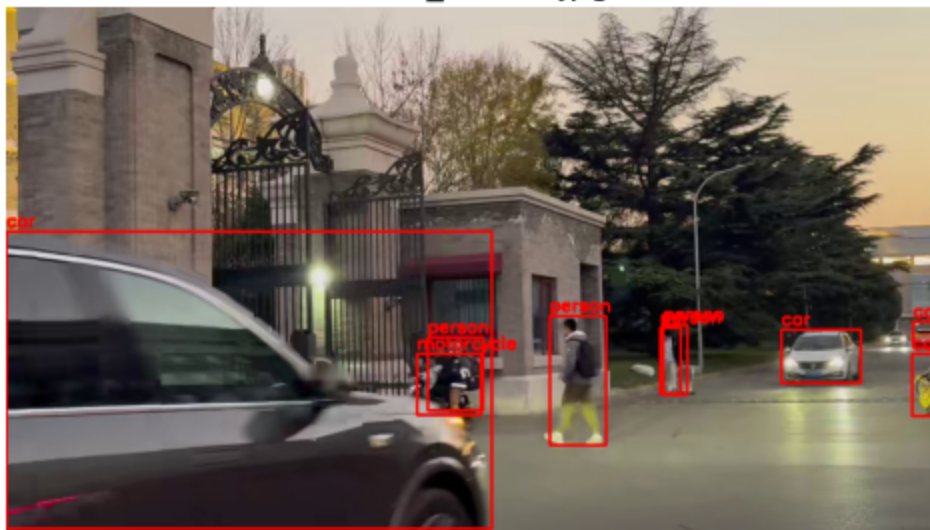
frame_02212.jpg



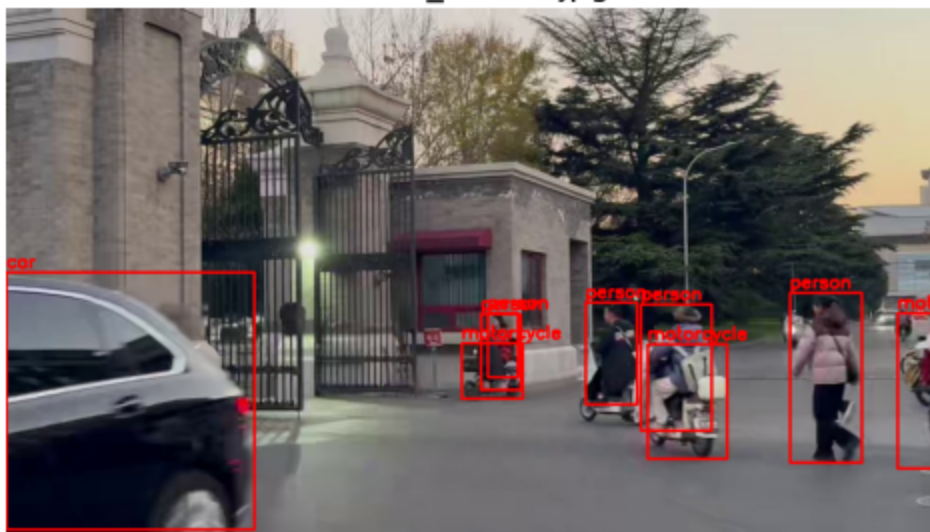
frame_01605.jpg



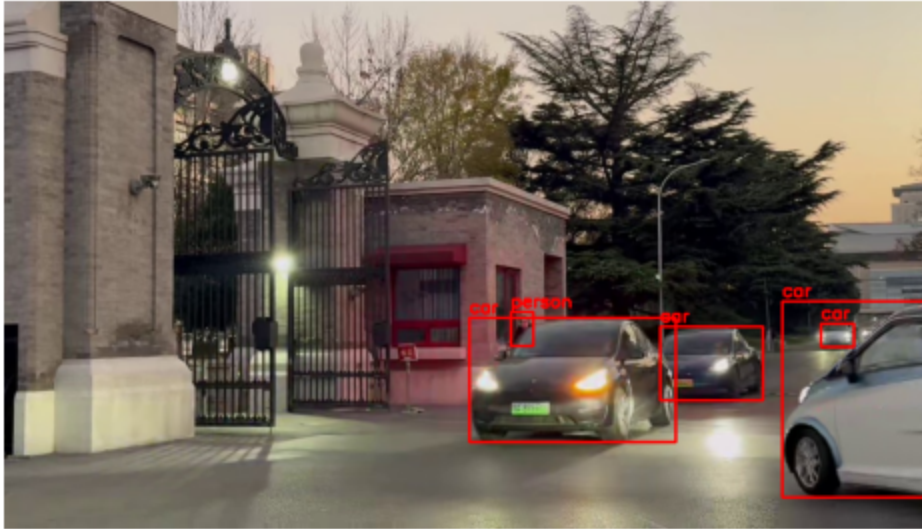
frame_02325.jpg



frame_00216.jpg



frame_01575.jpg



```
In [ ]: import subprocess
        yolov5_dir = "yolov5-master"

        subprocess.run([
            "python", f"{yolov5_dir}/train.py",
            "--weights", "yolov5n.pt",          # 训练YOLOv5n
            "--data", "data.yaml",              # 将data.yaml放在当前目录
            "--epochs", "50",
            "--img", "640",
            "--batch", "8",
            "--device", "0"                      # GPU 0
        ])

        print("🎉 训练完成!")
```

train: weights=yolov5n.pt, cfg=, data=data.yaml, hyp=yolov5-master/data/hyps/hyp.scratch-low.yaml, epochs=50, batch_size=8, imgsz=640, rect=False, resume=False, nosave=False, noval=False, noautoanchor=False, noplots=False, evolve=None, evolve_population=yolov5-master/data/hyps, resume_evolve=None, bucket=, cache=None, image_weights=False, device=0, multi_scale=False, single_cls=False, optimizer=SGD, sync_bn=False, workers=8, project=yolov5-master/runs/train, name=exp, exist_ok=False, quad=False, cos_lr=False, label_smoothing=0.0, patience=100, freeze=[0], save_period=-1, seed=0, local_rank=-1, entity=None, upload_dataset=False, bbox_interval=-1, artifact_alias=latest, ndjson_console=False, ndjson_file=False

YOLOv5 🚀 2025-11-23 Python-3.10.19 torch-2.9.1+cu128 CUDA:0 (NVIDIA RTX PRO 6000 Blackwell Workstation Edition, 9725 8MiB)

hyperparameters: lr=0.01, lrf=0.01, momentum=0.937, weight_decay=0.0005, warmup_epochs=3.0, warmup_momentum=0.8, warmup_bias_lr=0.1, box=0.05, cls=0.5, cls_pw=1.0, obj=1.0, obj_pw=1.0, iou_t=0.2, anchor_t=4.0, fl_gamma=0.0, hsv_h=0.015, hsv_s=0.7, hsv_v=0.4, degrees=0.0, translate=0.1, scale=0.5, shear=0.0, perspective=0.0, flipud=0.0, fliplr=0.5, mosaic=1.0, mixup=0.0, copy_paste=0.0

Comet: run 'pip install comet_ml' to automatically track and visualize YOLOv5 🚀 runs in Comet

TensorBoard: Start with 'tensorboard --logdir yolov5-master/runs/train', view at <http://localhost:6006/>
Overriding model.yaml nc=80 with nc=4

	from	n	params	module	arguments
0	-1	1	1760	models.common.Conv	[3, 16, 6, 2, 2]
1	-1	1	4672	models.common.Conv	[16, 32, 3, 2]
2	-1	1	4800	models.common.C3	[32, 32, 1]
3	-1	1	18560	models.common.Conv	[32, 64, 3, 2]
4	-1	2	29184	models.common.C3	[64, 64, 2]
5	-1	1	73984	models.common.Conv	[64, 128, 3, 2]
6	-1	3	156928	models.common.C3	[128, 128, 3]
7	-1	1	295424	models.common.Conv	[128, 256, 3, 2]
8	-1	1	296448	models.common.C3	[256, 256, 1]
9	-1	1	164608	models.common.SPPF	[256, 256, 5]
10	-1	1	33024	models.common.Conv	[256, 128, 1, 1]
11	-1	1	0	torch.nn.modules.upsampling.Upsample	[None, 2, 'nearest']
12	[-1, 6]	1	0	models.common.Concat	[1]
13	-1	1	90880	models.common.C3	[256, 128, 1, False]
14	-1	1	8320	models.common.Conv	[128, 64, 1, 1]
15	-1	1	0	torch.nn.modules.upsampling.Upsample	[None, 2, 'nearest']
16	[-1, 4]	1	0	models.common.Concat	[1]
17	-1	1	22912	models.common.C3	[128, 64, 1, False]
18	-1	1	36992	models.common.Conv	[64, 64, 3, 2]
19	[-1, 14]	1	0	models.common.Concat	[1]
20	-1	1	74496	models.common.C3	[128, 128, 1, False]
21	-1	1	147712	models.common.Conv	[128, 128, 3, 2]
22	[-1, 10]	1	0	models.common.Concat	[1]

```
23          -1  1    296448  models.common.C3          [256, 256, 1, False]
24    [17, 20, 23]  1    12177  models.yolo.Detect    [4, [[10, 13, 16, 30, 33, 23], [30, 61, 6
2, 45, 59, 119], [116, 90, 156, 198, 373, 326]], [64, 128, 256]]
Model summary: 214 layers, 1769329 parameters, 1769329 gradients, 4.2 GFLOPs
```

github: skipping check (not a git repository), for updates see <https://github.com/ultralytics/yolov5>


```

Transferred 343/349 items from yolov5n.pt
/workspace/yolov5-master/models/common.py:898: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp.autocast('cuda', args...)` instead.
    with amp.autocast(enabled=True):
/workspace/yolov5-master/models/common.py:898: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp.autocast('cuda', args...)` instead.
    with amp.autocast(enabled=True):
AMP: checks passed ✓
optimizer: SGD(lr=0.01) with parameter groups 57 weight(decay=0.0), 60 weight(decay=0.0005), 60 bias
train: Scanning /workspace/datasets/labels/train.cache... 1800 images, 1 backgrounds, 0 corrupt: 100%|██████████| 1800/1800 [00:00<?, ?it/s]
val: Scanning /workspace/datasets/labels/val.cache... 0 images, 2172 backgrounds, 0 corrupt: 100%|██████████| 2172/2172 [00:00<?, ?it/s]

AutoAnchor: 5.60 anchors/target, 1.000 Best Possible Recall (BPR). Current anchors are a good fit to dataset ✓
Plotting labels to yolov5-master/runs/train/exp3/labels.jpg...
/workspace/yolov5-master/train.py:357: FutureWarning: `torch.cuda.amp.GradScaler(args...)` is deprecated. Please use `torch.amp.GradScaler('cuda', args...)` instead.
    scaler = torch.cuda.amp.GradScaler(enabled=True)
Image sizes 640 train, 640 val
Using 8 dataloader workers
Logging results to yolov5-master/runs/train/exp3
Starting training for 50 epochs...

```

Epoch	GPU_mem	box_loss	obj_loss	cls_loss	Instances	Size
0%	0/225	[00:00<?, ?it/s]	/workspace/yolov5-master/train.py:414: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp.autocast('cuda', args...)` instead.			
with torch.cuda.amp.autocast(enabled=True):						
0/49	0.965G	0.1319	0.05083	0.05349	100	640: 0% 1/225 [00:00<01:48, 2.07it/s]
/workspace/yolov5-master/train.py:414: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp.autocast('cuda', args...)` instead.						
with torch.cuda.amp.autocast(enabled=True):						
0/49	0.973G	0.1323	0.04882	0.05406	94	640: 0% 1/225 [00:00<01:48, 2.07it/s]
/workspace/yolov5-master/train.py:414: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp.autocast('cuda', args...)` instead.						
with torch.cuda.amp.autocast(enabled=True):						
0/49	0.973G	0.1314	0.0481	0.0539	79	640: 1% 3/225 [00:00<00:36, 6.13it/s]
/workspace/yolov5-master/train.py:414: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp.autocast('cuda', args...)` instead.						
with torch.cuda.amp.autocast(enabled=True):						
0/49	0.973G	0.1316	0.04798	0.05401	90	640: 1% 3/225 [00:00<00:36, 6.13it/s]
/workspace/yolov5-master/train.py:414: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp.autocast('cuda', args...)` instead.						

```

36.46it/s]/workspace/yolov5-master/train.py:414: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp.autocast('cuda', args...)` instead.
  with torch.cuda.amp.autocast(amp):
    49/49      1.01G    0.02978    0.0393    0.006353      114      640: 100%|██████████| 224/225 [00:06<00:00,
37.04it/s]/workspace/yolov5-master/train.py:414: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp.autocast('cuda', args...)` instead.
  with torch.cuda.amp.autocast(amp):
    49/49      1.01G    0.02976    0.03923    0.006341      80      640: 100%|██████████| 225/225 [00:06<00:00,
35.88it/s]
      Class      Images  Instances      P      R      mAP50  mAP50-95: 100%|██████████| 136/136 [0
0:02<00:00, 50.03it/s]
      all      2172      0      0      0      0      0
WARNING ⚠ no labels found in val set, can not compute metrics without labels

50 epochs completed in 0.144 hours.
Optimizer stripped from yolov5-master/runs/train/exp3/weights/last.pt, 3.8MB
Optimizer stripped from yolov5-master/runs/train/exp3/weights/best.pt, 3.8MB

Validating yolov5-master/runs/train/exp3/weights/best.pt...
Fusing layers...
Model summary: 157 layers, 1764577 parameters, 0 gradients, 4.1 GFLOPs
      Class      Images  Instances      P      R      mAP50  mAP50-95: 100%|██████████| 136/136 [0
0:02<00:00, 48.70it/s]
      all      2172      0      0      0      0      0
WARNING ⚠ no labels found in val set, can not compute metrics without labels
Results saved to yolov5-master/runs/train/exp3
🎉 训练完成!

```

使用轻量级模型进行训练，从yolo5n获取参数权重来启动训练。

```

In [15]: !python yolov5-master/detect.py \
  --weights /workspace/yolov5-master/runs/train/exp3/weights/best.pt \
  --source /workspace/frames \
  --save-txt \
  --vid-stride 1

```

detect: weights=['/workspace/yolov5-master/runs/train/exp3/weights/best.pt'], source=/workspace/frames, data=yolov5-master/data/coco128.yaml, imgsz=[640, 640], conf_thres=0.25, iou_thres=0.45, max_det=1000, device=, view_img=False, save_txt=True, save_format=0, save_csv=False, save_conf=False, save_crop=False, nosave=False, classes=None, agnostic_nms=False, augment=False, visualize=False, update=False, project=yolov5-master/runs/detect, name=exp, exist_ok=False, line_thickness=3, hide_labels=False, hide_conf=False, half=False, dnn=False, vid_stride=1
YOLOv5 🚀 2025-11-23 Python-3.10.19 torch-2.9.1+cu128 CUDA:0 (NVIDIA RTX PRO 6000 Blackwell Workstation Edition, 97258MiB)

Fusing layers...

Model summary: 157 layers, 1764577 parameters, 0 gradients, 4.1 GFLOPs

image 1/3971 /workspace/frames/frame_00000.jpg: 384x640 6 persons, 1 bicycle, 1 car, 1 motorcycle, 10.2ms
image 2/3971 /workspace/frames/frame_00001.jpg: 384x640 5 persons, 1 bicycle, 1 car, 1 motorcycle, 2.4ms
image 3/3971 /workspace/frames/frame_00002.jpg: 384x640 4 persons, 1 bicycle, 1 car, 1 motorcycle, 1.9ms
image 4/3971 /workspace/frames/frame_00003.jpg: 384x640 3 persons, 1 bicycle, 1 car, 1 motorcycle, 1.9ms
image 5/3971 /workspace/frames/frame_00004.jpg: 384x640 5 persons, 1 bicycle, 1 car, 1 motorcycle, 1.9ms
image 6/3971 /workspace/frames/frame_00005.jpg: 384x640 5 persons, 1 bicycle, 1 car, 2 motorcycles, 1.9ms
image 7/3971 /workspace/frames/frame_00006.jpg: 384x640 5 persons, 1 bicycle, 2 cars, 2 motorcycles, 1.9ms
image 8/3971 /workspace/frames/frame_00007.jpg: 384x640 4 persons, 1 bicycle, 1 car, 3 motorcycles, 1.9ms
image 9/3971 /workspace/frames/frame_00008.jpg: 384x640 5 persons, 1 bicycle, 2 cars, 2 motorcycles, 1.9ms
image 10/3971 /workspace/frames/frame_00009.jpg: 384x640 5 persons, 1 bicycle, 1 car, 3 motorcycles, 1.9ms
image 11/3971 /workspace/frames/frame_00010.jpg: 384x640 4 persons, 1 bicycle, 1 car, 3 motorcycles, 1.9ms
image 12/3971 /workspace/frames/frame_00011.jpg: 384x640 4 persons, 1 bicycle, 2 motorcycles, 1.9ms
image 13/3971 /workspace/frames/frame_00012.jpg: 384x640 4 persons, 1 bicycle, 2 motorcycles, 1.9ms
image 14/3971 /workspace/frames/frame_00013.jpg: 384x640 4 persons, 1 bicycle, 2 motorcycles, 1.9ms
image 15/3971 /workspace/frames/frame_00014.jpg: 384x640 4 persons, 1 bicycle, 2 motorcycles, 1.9ms
image 16/3971 /workspace/frames/frame_00015.jpg: 384x640 4 persons, 1 bicycle, 2 motorcycles, 1.9ms
image 17/3971 /workspace/frames/frame_00016.jpg: 384x640 5 persons, 1 bicycle, 2 motorcycles, 1.9ms
image 18/3971 /workspace/frames/frame_00017.jpg: 384x640 4 persons, 1 bicycle, 3 motorcycles, 1.9ms
image 19/3971 /workspace/frames/frame_00018.jpg: 384x640 4 persons, 1 bicycle, 1 car, 3 motorcycles, 1.9ms
image 20/3971 /workspace/frames/frame_00019.jpg: 384x640 4 persons, 1 bicycle, 1 car, 4 motorcycles, 1.9ms
image 21/3971 /workspace/frames/frame_00020.jpg: 384x640 4 persons, 1 bicycle, 1 car, 4 motorcycles, 1.9ms
image 22/3971 /workspace/frames/frame_00021.jpg: 384x640 4 persons, 1 bicycle, 1 car, 4 motorcycles, 1.9ms
image 23/3971 /workspace/frames/frame_00022.jpg: 384x640 4 persons, 1 bicycle, 1 car, 5 motorcycles, 2.0ms
image 24/3971 /workspace/frames/frame_00023.jpg: 384x640 4 persons, 1 bicycle, 1 car, 4 motorcycles, 1.9ms
image 25/3971 /workspace/frames/frame_00024.jpg: 384x640 4 persons, 1 bicycle, 1 car, 4 motorcycles, 1.9ms
image 26/3971 /workspace/frames/frame_00025.jpg: 384x640 4 persons, 1 bicycle, 1 car, 4 motorcycles, 1.9ms
image 27/3971 /workspace/frames/frame_00026.jpg: 384x640 3 persons, 1 bicycle, 2 cars, 6 motorcycles, 1.9ms
image 28/3971 /workspace/frames/frame_00027.jpg: 384x640 2 persons, 1 bicycle, 2 cars, 4 motorcycles, 1.9ms
image 29/3971 /workspace/frames/frame_00028.jpg: 384x640 3 persons, 1 bicycle, 2 cars, 4 motorcycles, 2.0ms
image 30/3971 /workspace/frames/frame_00029.jpg: 384x640 2 persons, 1 bicycle, 2 cars, 3 motorcycles, 1.9ms
image 31/3971 /workspace/frames/frame_00030.jpg: 384x640 2 persons, 1 bicycle, 2 cars, 3 motorcycles, 2.0ms
image 32/3971 /workspace/frames/frame_00031.jpg: 384x640 2 persons, 1 bicycle, 2 cars, 3 motorcycles, 1.9ms

```
image 3939/3971 /workspace/frames/frame_03938.jpg: 384x640 3 persons, 1 bicycle, 1.8ms
image 3940/3971 /workspace/frames/frame_03939.jpg: 384x640 3 persons, 2 bicycles, 1.8ms
image 3941/3971 /workspace/frames/frame_03940.jpg: 384x640 5 persons, 1 motorcycle, 1.9ms
image 3942/3971 /workspace/frames/frame_03941.jpg: 384x640 4 persons, 1.8ms
image 3943/3971 /workspace/frames/frame_03942.jpg: 384x640 3 persons, 1 bicycle, 1.8ms
image 3944/3971 /workspace/frames/frame_03943.jpg: 384x640 3 persons, 1 bicycle, 1.8ms
image 3945/3971 /workspace/frames/frame_03944.jpg: 384x640 4 persons, 1 bicycle, 1.9ms
image 3946/3971 /workspace/frames/frame_03945.jpg: 384x640 5 persons, 1 bicycle, 1 motorcycle, 1.8ms
image 3947/3971 /workspace/frames/frame_03946.jpg: 384x640 5 persons, 1 bicycle, 1 car, 1 motorcycle, 1.8ms
image 3948/3971 /workspace/frames/frame_03947.jpg: 384x640 3 persons, 1 bicycle, 2 cars, 1 motorcycle, 1.8ms
image 3949/3971 /workspace/frames/frame_03948.jpg: 384x640 4 persons, 2 bicycles, 1 car, 1 motorcycle, 1.8ms
image 3950/3971 /workspace/frames/frame_03949.jpg: 384x640 4 persons, 2 bicycles, 1 motorcycle, 1.8ms
image 3951/3971 /workspace/frames/frame_03950.jpg: 384x640 5 persons, 2 bicycles, 1 motorcycle, 1.8ms
image 3952/3971 /workspace/frames/frame_03951.jpg: 384x640 6 persons, 2 bicycles, 1 motorcycle, 1.8ms
image 3953/3971 /workspace/frames/frame_03952.jpg: 384x640 7 persons, 2 bicycles, 2 motorcycles, 1.8ms
image 3954/3971 /workspace/frames/frame_03953.jpg: 384x640 5 persons, 2 bicycles, 1 motorcycle, 1.8ms
image 3955/3971 /workspace/frames/frame_03954.jpg: 384x640 5 persons, 2 bicycles, 1.8ms
image 3956/3971 /workspace/frames/frame_03955.jpg: 384x640 4 persons, 2 bicycles, 1 motorcycle, 1.8ms
image 3957/3971 /workspace/frames/frame_03956.jpg: 384x640 4 persons, 2 bicycles, 1.8ms
image 3958/3971 /workspace/frames/frame_03957.jpg: 384x640 4 persons, 2 bicycles, 1.8ms
image 3959/3971 /workspace/frames/frame_03958.jpg: 384x640 3 persons, 2 bicycles, 1 car, 1.8ms
image 3960/3971 /workspace/frames/frame_03959.jpg: 384x640 4 persons, 1 bicycle, 1 motorcycle, 1.8ms
image 3961/3971 /workspace/frames/frame_03960.jpg: 384x640 5 persons, 1 bicycle, 1 motorcycle, 1.8ms
image 3962/3971 /workspace/frames/frame_03961.jpg: 384x640 5 persons, 1 bicycle, 1 motorcycle, 1.9ms
image 3963/3971 /workspace/frames/frame_03962.jpg: 384x640 5 persons, 2 bicycles, 1 motorcycle, 1.8ms
image 3964/3971 /workspace/frames/frame_03963.jpg: 384x640 7 persons, 2 bicycles, 2 motorcycles, 1.8ms
image 3965/3971 /workspace/frames/frame_03964.jpg: 384x640 7 persons, 2 bicycles, 2 motorcycles, 1.8ms
image 3966/3971 /workspace/frames/frame_03965.jpg: 384x640 6 persons, 2 bicycles, 1 motorcycle, 1.8ms
image 3967/3971 /workspace/frames/frame_03966.jpg: 384x640 7 persons, 2 bicycles, 1 motorcycle, 1.9ms
image 3968/3971 /workspace/frames/frame_03967.jpg: 384x640 6 persons, 2 bicycles, 1 motorcycle, 1.8ms
image 3969/3971 /workspace/frames/frame_03968.jpg: 384x640 7 persons, 2 bicycles, 1 motorcycle, 1.9ms
image 3970/3971 /workspace/frames/frame_03969.jpg: 384x640 6 persons, 2 bicycles, 2 motorcycles, 1.8ms
image 3971/3971 /workspace/frames/frame_03970.jpg: 384x640 6 persons, 2 bicycles, 2 motorcycles, 1.8ms
Speed: 0.1ms pre-process, 1.9ms inference, 0.4ms NMS per image at shape (1, 3, 640, 640)
Results saved to yolov5-master/runs/detect/exp3
3970 labels saved to yolov5-master/runs/detect/exp3/labels
```

推理时间稳定在1.8-1.9s左右，训练相对高效。

```
In [14]: import pandas as pd
```

```

import matplotlib.pyplot as plt

csv_path = "yolov5-master/runs/train/exp3/results.csv"
df = pd.read_csv(csv_path)

df.columns = df.columns.str.strip()

print("清理后的列名: ")
print(df.columns)

def plot_single(df, col, title):
    if col not in df.columns:
        print(f"[列不存在]: {col}")
        return
    plt.figure(figsize=(6,4))
    plt.plot(df[col], marker='o')
    plt.title(title)
    plt.grid(True)
    plt.show()

plot_single(df, "train/box_loss", "train/box_loss")
plot_single(df, "train/obj_loss", "train/obj_loss")
plot_single(df, "train/cls_loss", "train/cls_loss")

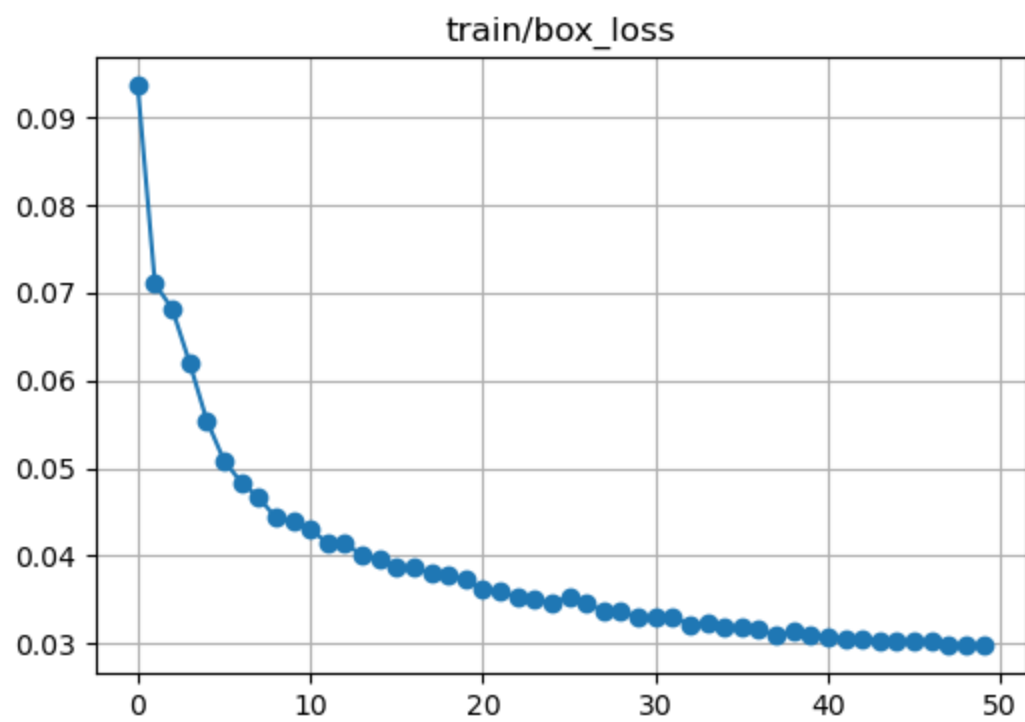
```

清理后的列名:

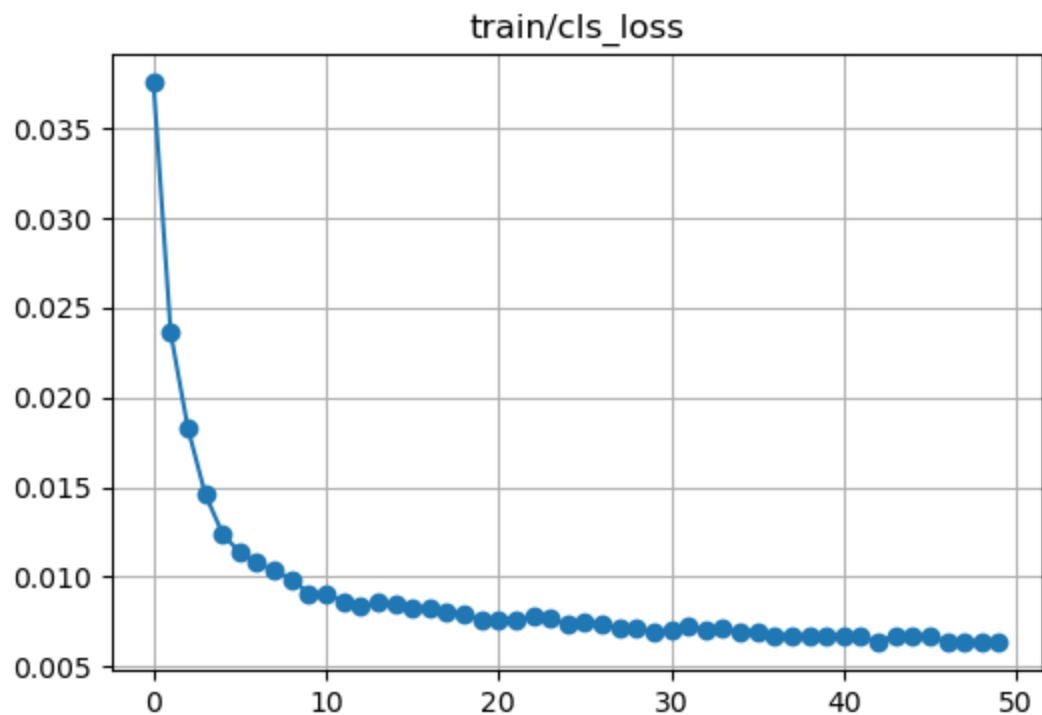
```

Index(['epoch', 'train/box_loss', 'train/obj_loss', 'train/cls_loss',
      'metrics/precision', 'metrics/recall', 'metrics/mAP_0.5',
      'metrics/mAP_0.5:0.95', 'val/box_loss', 'val/obj_loss', 'val/cls_loss',
      'x/lr0', 'x/lr1', 'x/lr2'],
      dtype='object')

```







train/box loss曲线持续下降到约0.03，模型在目标边界框回归任务上的误差逐渐减小，定位精度持续提升。

train/obj_loss曲线波动下降到约0.04，模型判断准确率较高，预测的置信度高

train/cls loss曲线快速下降到0.01以下，模型分类正确率大幅度提高，类别识别精度显著提高。

综合来看，模型收敛，学习效果较好。

In [16]: `import cv2`


```

import os
from glob import glob

# detect.py 输出目录
pred_dir = 'yolov5-master/runs/detect/exp3'
out_video = 'val_predicted_from_detect.mp4'

# 读取预测后的图片
images = sorted(
    glob(os.path.join(pred_dir, "*.jpg")) +
    glob(os.path.join(pred_dir, "*.png"))
)

# 读取第一张图片确定尺寸
frame = cv2.imread(images[0])
h, w, _ = frame.shape

fps = 24
fourcc = cv2.VideoWriter_fourcc(*'mp4v')
out = cv2.VideoWriter(out_video, fourcc, fps, (w, h))

# 合成视频
for img_path in images:
    frame = cv2.imread(img_path)
    out.write(frame)

out.release()

print("已根据 detect.py 预测结果生成视频:", out_video)

```

已根据 detect.py 预测结果生成视频: val_predicted_from_detect.mp4

读取目标检测后生成的所有图像并按顺序拼接为视频

由于训练和预测过程中log过多（共占约1000页），故而在保存为pdf后删除了中间部分所占页面